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01 Executive Overview

Quetta Crave is an independent gourmet fast-food restaurant located on Circular Road in Rahim Colony, Quetta (near GOGO Pizza). To address severe margin erosion from third-party delivery aggregators and eliminate order entry mistakes during peak dinner hours, we designed, developed, and deployed an in-house digital ordering web platform coupled with a normalized MySQL database. This report provides a practical overview of the business problem, the engineering architecture delivered, and how agentic AI tools were used to accelerate development.

02 The Business Problem

Local food businesses in Quetta operate in a challenging environment characterized by low credit card usage, fluctuating ingredient costs, and high platform commissions:

1. 25%–30% Aggregator Commissions

Third-party delivery platforms take nearly a third of order revenue. For prime beef burgers and imported cheese pizzas with tight margins, third-party delivery was financially unsustainable.

2. WhatsApp & Phone Ordering Chaos

During evening rushes (8 PM to 1 AM), manual phone calls and WhatsApp audio notes caused missed calls, wrong custom burger orders, incorrect street delivery addresses, and customer frustration.

3. Local Payment Method Incompatibility

Over 90% of customers in Quetta do not use Visa/Mastercard. Conventional checkouts (Stripe/PayPal) fail completely. Customers need *EasyPaisa*, *JazzCash*, *SadaPay*, *NayaPay*, and *Cash on Delivery*.

4. Price Volatility & Menu Stagnation

Dairy and meat commodity prices fluctuate rapidly in Balochistan. Printed menus and static websites created lag, leading to sold-out dishes being ordered or meals sold at outdated pricing.

Primary Mission: Deliver a lightweight, zero-maintenance digital storefront with native Pakistani mobile wallet integration, an automated free-delivery threshold engine (Orders ≥ Rs. 1,000), a secure owner admin portal, and a normalized SQL backend — hosted 24/7 with zero hosting subscription fees.

Problem Breakdown & Technical Mitigations

Identified Challenge	Direct Business Impact	Delivered Technical Solution
High Aggregator Fees	Lost ~30% margin per delivery order	Self-hosted web platform deployed on GitHub Pages with 0% commission
Manual Order Entry Errors	Kitchen remakes and delivery delays	Interactive cart drawer with exact itemization, quantity controls & address capture
Payment Friction	Riders stranded without change at doorstep	EasyPaisa (Till #88291), JazzCash (Till #44102), SadaPay & COD selector
Delayed Menu Updates	Selling out-of-stock or mispriced food	Password-protected in-browser Admin CRUD portal for instant dish management

03 Architecture & Technical Solution

We architected an ultra-lean, high-performance web application designed to load instantly on 3G/4G cellular networks across Quetta, backed by a persistent relational data model.

Responsive Client UI

Semantic HTML5, custom CSS design system (Outfit & Poppins), 3D glowing brand badges, staggered floating product cards, and FontAwesome icons.

Smart Cart & Upsell Engine

Dynamic subtotal computation with automated **Free Delivery** unlocking at Rs. 1,000 threshold (Rs. 150 standard fee for smaller orders) via visual progress tracker.

Secure Admin Management

Protected by credentials (**CRAVELODGE** / **foodfanatic007**). Real-time CRUD for dishes, live orders dashboard, customer review moderation, and photo uploads.

1. User Experience & Localized Checkout

- 3D Pop-Up Brand Identity:** Animated glowing header with interactive hover card displaying exact landmark navigation (*Rahim Colony near GOGO Pizza, Circular Road*).
- 25+ Catalog Dishes in 5 Categories:** Gourmet Smash Burgers, Artisan Cheesy Pizzas, Crispy Broast & Wings, Shawarmas & Wraps, Loaded Fries & Freakshakes.
- Pakistani Payment Gateways:** Cash on Delivery, EasyPaisa (Till ID: **88291**), JazzCash (Till ID: **44102**), and Meezan Bank / SadaPay with Transaction ID verification.
- 5-Star Customer Testimonials:** Dynamic customer review cards and interactive review submission modal saving directly to persistent storage.

2. Relational Database Architecture (**database.sql**)

A standalone MySQL 8.0 schema was engineered with 7 normalized tables, constraints, analytical views, and stored procedures:

Table / Entity	Primary Purpose & Relational Constraints	Key Fields
categories	Menu category taxonomy with slug, icons & display sort order	id, name, slug, icon, display_order
products	25 pre-seeded menu items linked to categories (ON DELETE CASCADE)	id, category_id, name, price, badge, rating, is_available
orders	Customer orders with Quetta address, payment option & status	id, order_number, customer_phone, subtotal, delivery_fee, status
order_items	Itemized line records linked to orders with cascading integrity	id, order_id, product_id, price, quantity, subtotal
reviews	5-star customer ratings with moderation approval flag	id, customer_name, rating, comment, favorite_dish, is_approved
admin_users	Administrative authentication table with salted password hashes	id, username, password_hash, full_name, role

Stored Procedure **sp_create_order** : Embeds core pricing business logic directly in the database engine — calculating order subtotal; if ≥ Rs. 1,000, sets delivery fee to Rs. 0.00; otherwise applies the standard Rs. 150.00 fee automatically.

04 AI Tools Used During Development

AI was employed as an active pair-programming collaborator throughout the design, coding, and deployment lifecycle:

1. Antigravity AI Agent (Gemini 3.7 Flash)

Served as the primary architectural co-pilot. Generated semantic HTML/CSS design tokens, cart state management, and 3D floating keyframe physics.

2. SQL Schema & Relational Modeling AI

Synthesized normalized MySQL 8.0 DDL scripts, foreign keys, stored procedure `sp_create_order`, and analytical views (`v_popular_dishes`).

3. Automated CI/CD & GitHub REST API Integration

Orchestrated automated Git commits, base64 payload serialization, repository configuration, and GitHub Pages build verification via REST endpoints without requiring local Git installs.

4. UX & Copy Localization AI

Crafted authentic culinary copy tailored specifically to Quetta foodies (smash beef patties, malai boti, toum garlic shawarma) and accurately structured local payment instructions.

05 Verification & Production Deployment

The platform was fully built, verified, and pushed to GitHub with 24/7 continuous availability:

Production Status & Availability:

- Live Website (24/7 Uptime): <https://asadibrahim121.github.io/quetta-crave/>
- GitHub Repository: <https://github.com/AsadIbrahim121/quetta-crave>
- Hosting Infrastructure: GitHub Pages backed by Fastly global CDN edge nodes with automated HTTPS SSL.
- Verification Result: HTTP 200 OK • Response Time < 150ms • Zero external server maintenance cost.

06 Summary & Next Steps

By combining modern client-side engineering with localized payment options and an AI-accelerated development cycle, Quetta Crave reclaimed its profit margins and established a polished digital presence. Upcoming enhancements include automated SMS dispatch to kitchen tablets and direct WhatsApp webhook notifications for delivery riders.

Report Prepared & Authenticated By:

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System Deployment Status:

✓ DEPLOYED & LIVE 24/7